Supplement: Dynamic Intra-Cell Repositioning in Free-Floating Bike Sharing Systems Using Approximate Dynamic Programming

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Appendix A: Details of test instances

In the numerical experiments in Section 5, we construct the test instances based on the real data set of a major free-floating bike sharing company in China (Mobike 2017). In this appendix, we first provide an overview of the data set and describe how we preprocess the data in Appendix A.1. Then, we present the estimation of the bike supply and demand rates at each node (Appendix A.2) and the initial bike allocation (Appendix A.3).

A.1. Data overview and preprocessing

The original data set of Mobike (2017) includes 3,214,096 bike trips in Beijing over 14 days, involving 349,693 users and 485,465 bikes. As shown in Figure 1, each bike trip has its start time, start location, and end location. In our numerical experiments, we focus on the bike usage patterns on weekdays. After removing the weekend days and the days with incomplete trip data, we have a data set of 2,004,942 bike trips over seven days, involving 320,457 users and 452,008 bikes.

	orderid	userid	bikeid	starttime	geohashed_start_loc	geohashed_end_loc
0	1893973	451147	210617	2017-05-14 22:16:50	wx4snhx	wx4snhj
1	4657992	1061133	465394	2017-05-14 22:16:52	wx4dr59	wx4dquz
2	2965085	549189	310572	2017-05-14 22:16:51	wx4fgur	wx4fu5n
3	4548579	489720	456688	2017-05-14 22:16:51	wx4d5r5	wx4d5r4
4	3936364	467449	403224	2017-05-14 22:16:50	wx4g27p	wx4g266

Figure 1 Examples of bike trips in the original data set (Mobike 2017)

As shown in Figure 1, the start and end locations are encoded by Geohash. That is, Beijing is divided into grids with a side length of around 150 meters, and all the start and end locations are located at the

centers of the grids (Gao et al. 2018, Zhou et al. 2021), referred to as "discrete locations." For example, in Figure 2, we display the discrete locations (represented by the green dots) within and around the cell in the base instances.

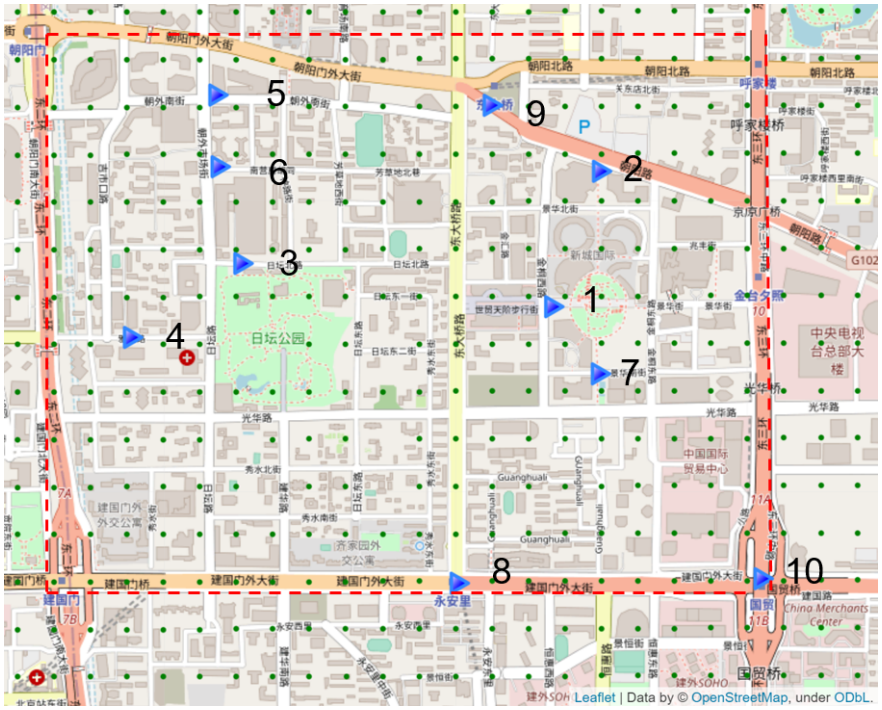


Figure 2 **Discrete locations and gathering points within and around the cell**

As described in Section 5.1, we select nodes (gathering points) 1 – 10 within the cell, represented by the blue triangles in Figure 2. We need to associate each node with its nearby discrete locations and aggregate their bike arrivals and departures to the node. If a discrete location is within $\zeta = 300$ meters from one node, we assign the discrete location to the node. If a discrete location is within ζ meters from more than one node, we assign the discrete location to the nearest node. Note that for a node near the boundary of the cell (e.g., node 10), we also consider its discrete locations (within ζ meters) outside the cell when aggregating the bike arrivals and departures. Note also that all the distances in this section are measured in Manhattan distance.

A.2. Estimation of bike supply and demand rates

In our numerical experiments in Section 5.1, we represent the bike supply and demand at each node $i \in \mathcal{N}$ or on each arc $(i, j) \in \mathcal{A}$ during each time interval $e \in \mathcal{E}$ as independent Poisson processes. In this section, we describe how we estimate the bike supply and demand rates based on Mobike (2017) and the corresponding goodness-of-fit tests.

For each node $i \in \mathcal{N}$ during each time interval $e \in \mathcal{E}$, we obtain a set of bike *arrivals* and the corresponding set of interarrival times between bike arrivals from Mobike (2017). We estimate the bike *supply* rate λ_{ei}^{+} based on the ratio of the total bike arrivals to their total time duration, and then run a Kolmogorov-Smirnov

(K-S) goodness-of-fit test to check whether the empirical distribution of the interarrival times in the set fits the exponential distribution with parameter $1/\lambda_{ei}^{+}$. (Interested readers can refer to Section 10.6 of DeGroot and Schervish (2010) for more details about the K-S test.) We estimate and test the bike *demand* rate λ_{ei}^{-} similarly, using the data of bike *departures* from Mobike (2017).

In the base instances, we need to estimate λ_{ei}^{+} and λ_{ei}^{-} for each of the 10 nodes in $\mathcal{N}$ and each of the 5 time intervals in $\mathcal{E}$. In only three out of the $10\times5\times2=100$ K-S tests, the null hypothesis is rejected at the significance level of 0.05. Therefore, we can reasonably approximate the bike supply and demand at each node during each time interval as independent Poisson processes, which is also in line with the industry practice. We make a few minor adjustments on the estimated bike supply and demand rates according to expert opinions to better reveal the demand patterns under study.

A.3. Initial bike allocation

We determine the initial bike allocation as in Figure 3. Let N^0 denote the total number of bikes in the initial network, i.e., $N^0=\sum_{i\in\mathcal{N}'} n_{0i}+\sum_{(i,j)\in\mathcal{A}} n_{0ij}$. In the base instances, we set $N^0=300$, based on the data set of Mobike (2017) and the practice of a major free-floating bike sharing company in China.

As described in Section 5.1, in the initial network, we allocate $N^{0,N}=(1-\psi^A)N^0$ bikes to the nodes and $N^{0,A}=\psi^A N^0$ bikes to the arcs, where ψ^A is the proportion of bikes on the arcs. In the base instances, we set $\psi^A=50\%$ and $N^{0,N}=N^{0,A}=150$. Then, we follow the steps below to allocate bikes to each node and arc. Note that the initial number of bikes is set to zero at the depot, i.e., $n_{00}=0$.

1. Allocate $\psi^- N^{0,N}$ bikes to the evening-peak-*demand* nodes and the rest $(1-\psi^-)N^{0,N}$ bikes to the evening-peak-*supply* nodes, where $\psi^-=0.8$.

   (a) At evening-peak-demand node i, the initial number of bikes n_{0i} is set proportional to its expected net demand during the working time $\sum_{e\in\mathcal{E}}(\lambda_{ei}^{-}-\lambda_{ei}^{+})$.

   (b) At evening-peak-supply node i, the initial number of bikes n_{0i} is set proportional to the reciprocal of its expected net demand during the working time $\frac{1}{\sum_{e\in\mathcal{E}}(\lambda_{ei}^{-}-\lambda_{ei}^{+})}$.

2. On arc $(i,j)\in\mathcal{A}$, the initial number of bikes n_{0ij} is set proportional to the initial number of bikes at its endpoint nodes, i.e., $n_{0i}+n_{0j}$.

Appendix B: Impact of stochastic travel time

In this paper, we assume deterministic travel time of τ_{ij} on arc $(i,j)\in\mathcal{A}$. In the following, we perform an exploratory study on the impact of stochastic travel time. Specifically, we add a perturbation $\epsilon_{ij}\sim N(0,\sigma_{ij})$ on the travel time of τ_{ij} on each arc $(i,j)\in\mathcal{A}$. We experiment on various settings of σ_{ij}, that is, $\{10\%\tau_{ij},20\%\tau_{ij},30\%\tau_{ij}\}$. In all the experiments, we keep the policy parameters the same as the deterministic travel time settings in the base instances (see Table 5), and evaluate these policies under the stochastic travel time settings. Note that the performance of NRP (that is, the "do nothing policy") is not affected by the stochastic travel time.

In Table 1, we show the demand satisfaction ratio of each policy (averaged over the 200 replications) under deterministic travel time (column (1)), under stochastic travel time with different σ_{ij} (columns (2) – (4)), and their differences (columns (5) – (7)). We observe that the impact of stochastic travel time is limited for

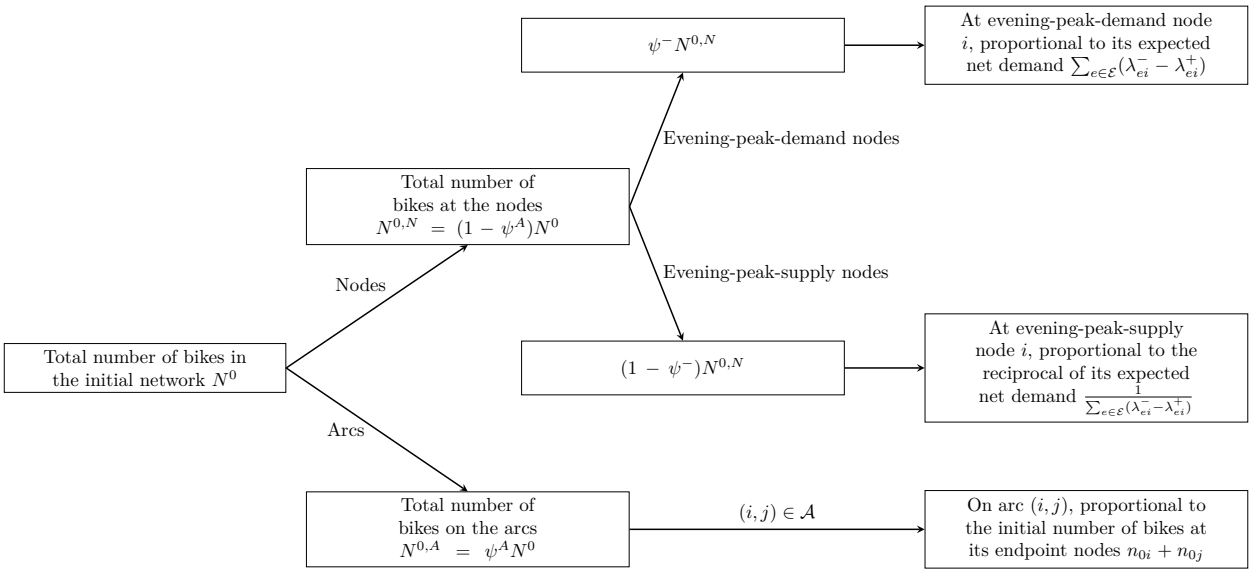


Figure 3 Initial bike allocation

each policy. With $\sigma_{ij}=10\%\tau_{ij}$, the differences between the demand satisfaction ratios under deterministic and stochastic travel time are almost negligible. With $\sigma_{ij}=30\%\tau_{ij}$, which is already significant in regard to the travel time stochasticity of e-tricycles, the differences are within 1%. Further, we observe that the relative performances among the policies under stochastic travel time remain the same as those under deterministic travel time.

Table 1 Demand satisfaction ratios under deterministic and stochastic travel time

	Deterministic travel time (1)	$\sigma_{ij}=0.1\tau_{ij}$ (2)	$\sigma_{ij}=0.2\tau_{ij}$ (3)	$\sigma_{ij}=0.3\tau_{ij}$ (4)	(1) − (2) (5)	(1) − (3) (6)	(1) − (4) (7)
NRP	77.3%	-	-	-	-	-	-
STR	81.1%	81.0%	80.9%	80.3%	0.1%	0.2%	0.8%
GLA	85.0%	85.0%	84.9%	84.5%	0.0%	0.1%	0.5%
PFA	91.2%	91.2%	91.1%	90.3%	0.0%	0.1%	0.9%

References

DeGroot MH, Schervish MJ, 2010 *Probability and Statistics 4th edition* (Addison-Wesley).

Gao Y, Song C, Shu H, Pei T, 2018 *Spatial-temporal characteristics of source and sink points of mobikes in beijing and its scheduling strategy. Journal of Geo-information Science* 20(8):1123–1138.

Mobike, 2017 *Mobike data mining competition at biendata*. https://biendata.com/competition/mobike/, in Chinese, last accessed on 14 February 2021.

Zhou Y, Zhao M, Lam WH, Chen A, Sze N, Tang S, Chen Y, Zhang T, 2021 *Modeling the impact of dock-less bike-sharing system on outpatient trips. Sustainable Cities and Society* 69:102853.